END-TO-END DATA ANALYTICS PROJECT

Multi-Branch Retail Performance & Operations Intelligence

Excel • MySQL • SQL • Power BI • DAX

Mumbai | Pune | Delhi | Bangalore

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OVERVIEW

Executive Summary

This initiative establishes a comprehensive operational intelligence framework for retail enterprises across four major urban centers, transforming raw retail data into an interactive BI solution covering sales, store performance, products, customers, shipments, returns and refunds.

- Initial profiling in Excel, structured analysis and validation in MySQL/SQL.
- Power BI + DAX for data modeling, KPI development and interactive reporting.
- Five management-ready dashboards, each independently validated against SQL.
- Findings translated into prioritized business recommendations.

$3.83B Gross Sales	**29.77%** YoY Growth
₹12.76K AOV	**50K** Customers
98.56% Repeat Rate	**5** Dashboards

BUSINESS CONTEXT

Business Problem & Objectives

BUSINESS PROBLEM

- Identify strongest and weakest cities/stores.
- Measure sales, orders, AOV and growth.
- Identify high-performing products and categories.
- Understand customer frequency, value and repeat behavior.
- Monitor shipment status and late delivery.
- Measure returns and refund value.

KEY OBJECTIVES

- Compare Mumbai, Pune, Delhi & Bangalore performance.
- Identify top products/categories and sales contribution.
- Analyze customer value, frequency and repeat behavior.
- Evaluate shipment performance and late-delivery patterns.
- Analyze returns/refunds for operational impact areas.
- Deliver validated, management-ready dashboards.

Dataset Overview

A 12-table retail data warehouse containing approximately 1.59 million records.

Table	Records
Categories	30
Customers	50,000
Employee	1,000
Order Items	600,000
Orders	300,000
Payments	300,000
Products	10,000
Promotions	50
Returns	30,000
Shipments	300,000
Stores	100
Suppliers	200

TOTAL RECORDS

1,591,380

CORE ANALYTICAL FLOW

- Customers → Orders → Order Items → Returns
- Orders → Shipments

Relationship and filter propagation were tested during development, particularly for city-level shipment and return analysis.

METHODOLOGY

Technology Stack & Analytics Workflow

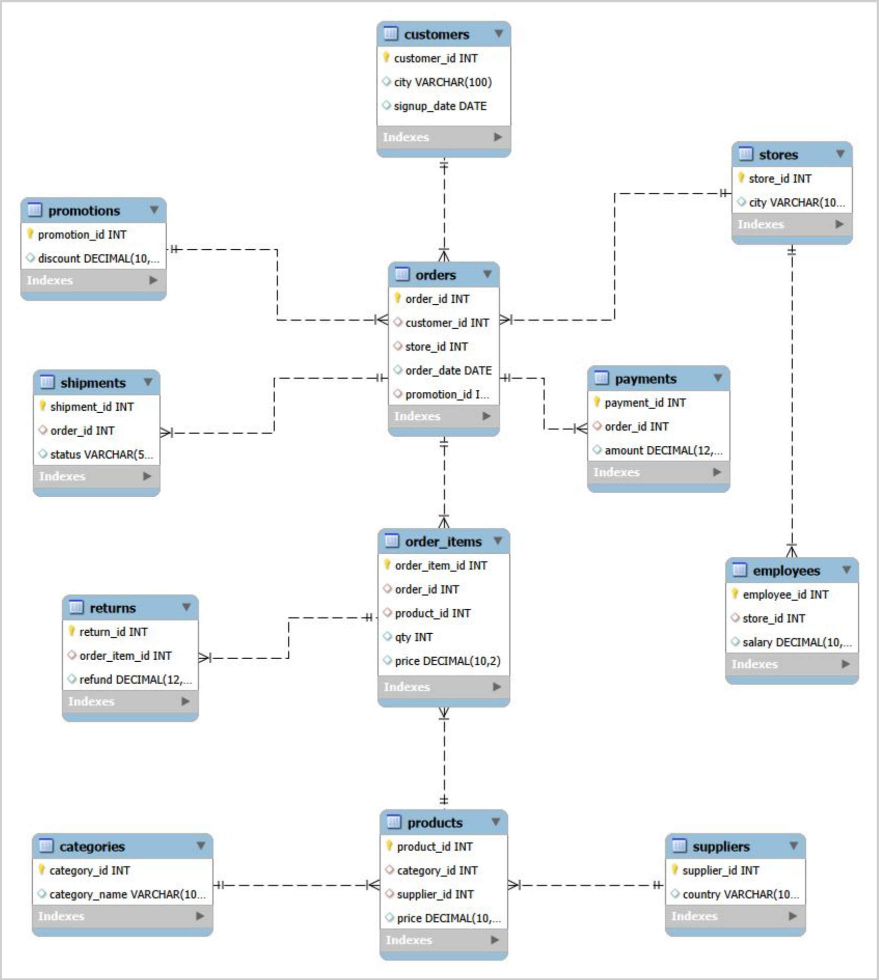
Excel	Initial data exploration
MySQL 8.0	Data warehouse querying, EDA, aggregation & validation
SQL	EDA, joins, aggregations, distinct counts, KPI validation
Power BI	Data modeling, relationships, slicers & dashboards
DAX	Reusable measures, ratios & calculated columns
Google Docs / Slides	Documentation & presentation
GitHub	Portfolio hosting & version control

ANALYTICS WORKFLOW

1. Raw Retail Data
2. Initial Data Inspection
3. MySQL/SQL Analysis
4. SQL KPI Validation
5. Power BI Data Model
6. DAX Measures
7. Dashboards
8. SQL Reconciliation
9. Business Insights
10. Recommendations

DATA ARCHITECTURE

Data Model



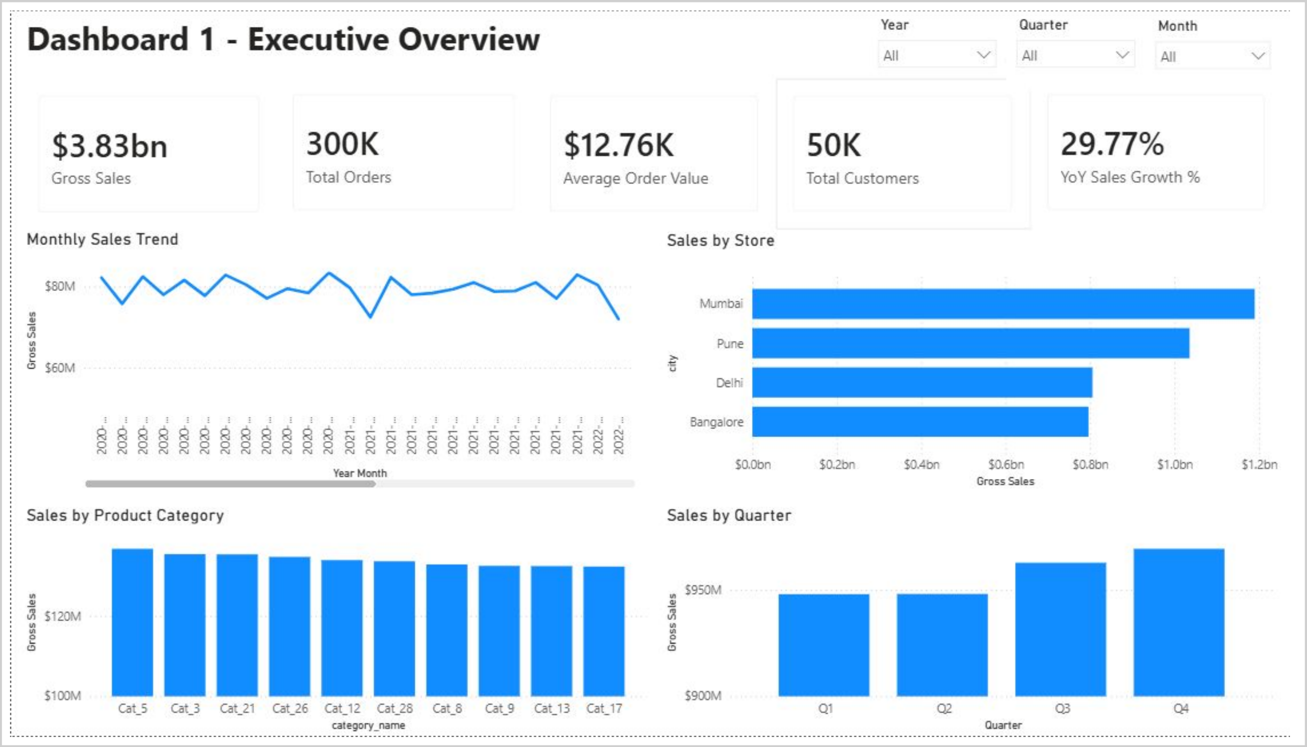
CORE ANALYTICAL FLOW

- Customers → Orders → Order Items → Returns
- Orders → Shipments

12 interconnected tables spanning customers, stores, employees, orders, order items, payments, shipments, returns, products, categories, suppliers and promotions.

Relationship and filter propagation were tested during development, particularly for city-level shipment and return analysis.

Executive Overview



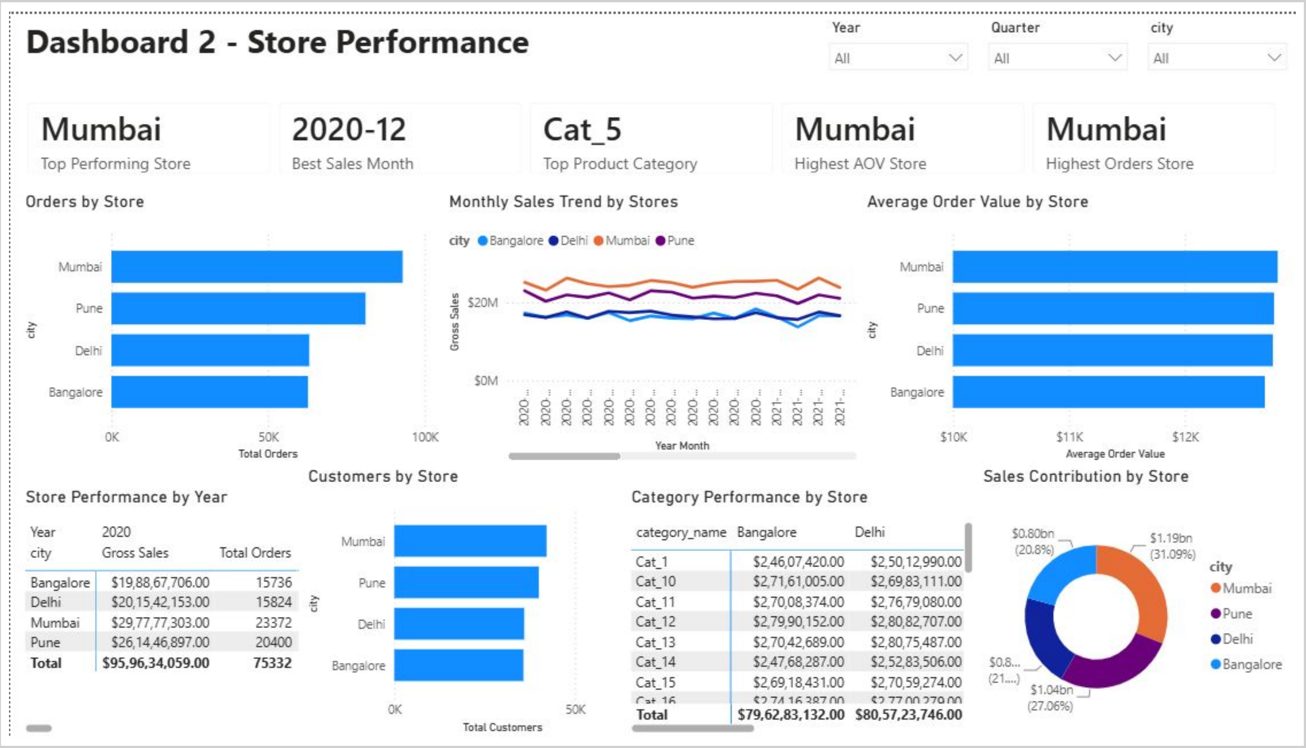
KEY FINDINGS

- Gross sales are approximately ₹3.83B.
- YoY sales growth is 29.77%.
- AOV is approximately ₹12.76K.
- Mumbai is the strongest revenue market.
- Q4 is the strongest quarter.

MANAGEMENT CONCLUSION

The retail business generated approximately $3.83B in gross sales across four cities, with Mumbai contributing the highest sales. Sales grew 29.77% YoY, while AOV reached ₹12.76K. Revenue remained relatively stable across months, with Q4 emerging as the strongest quarter.

Store Performance



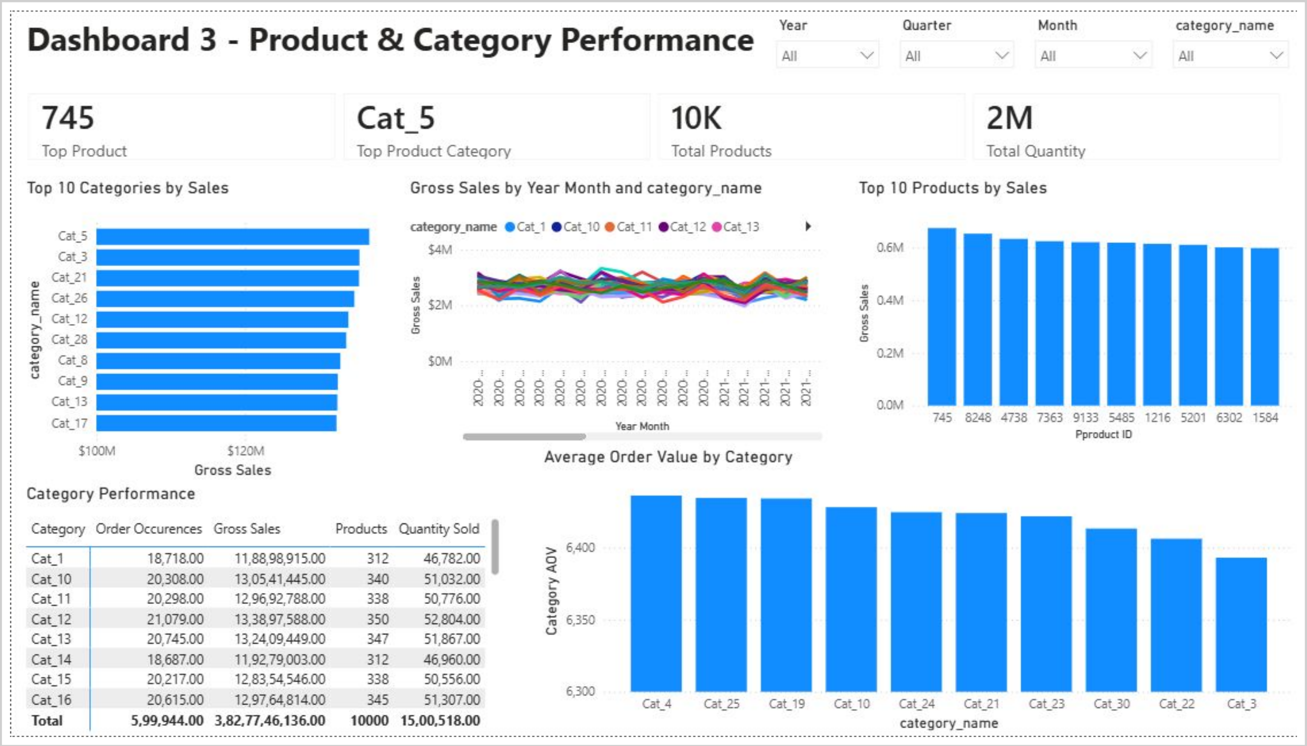
KEY FINDINGS

- Mumbai is the strongest-performing market.
- Mumbai leads order volume and customer base.
- Mumbai has the strongest AOV position.
- Mumbai contributes approximately 31% of overall sales.
- Category performance varies by city.

MANAGEMENT CONCLUSION

Mumbai leads in orders, customers, average order value and overall sales contribution. Pune follows, while Delhi and Bangalore show comparatively lower performance. Category-level performance also varies between cities, highlighting the need for location-specific strategies.

Product & Category Performance



Category	Order Occurences	Gross Sales	Products	Quantity Sold
Cat_1	18,718.00	11,88,98,915.00	312	46,782.00
Cat_10	20,308.00	13,05,41,445.00	340	51,032.00
Cat_11	20,298.00	12,96,92,788.00	338	50,776.00
Cat_12	21,079.00	13,38,97,588.00	350	52,804.00
Cat_13	20,745.00	13,24,09,449.00	347	51,867.00
Cat_14	18,687.00	11,92,79,003.00	312	46,960.00
Cat_15	20,217.00	12,83,54,546.00	338	50,556.00
Cat_16	20,615.00	12,97,64,814.00	345	51,307.00
Total	**5,99,944.00**	**3,82,77,46,136.00**	**10000**	**15,00,518.00**

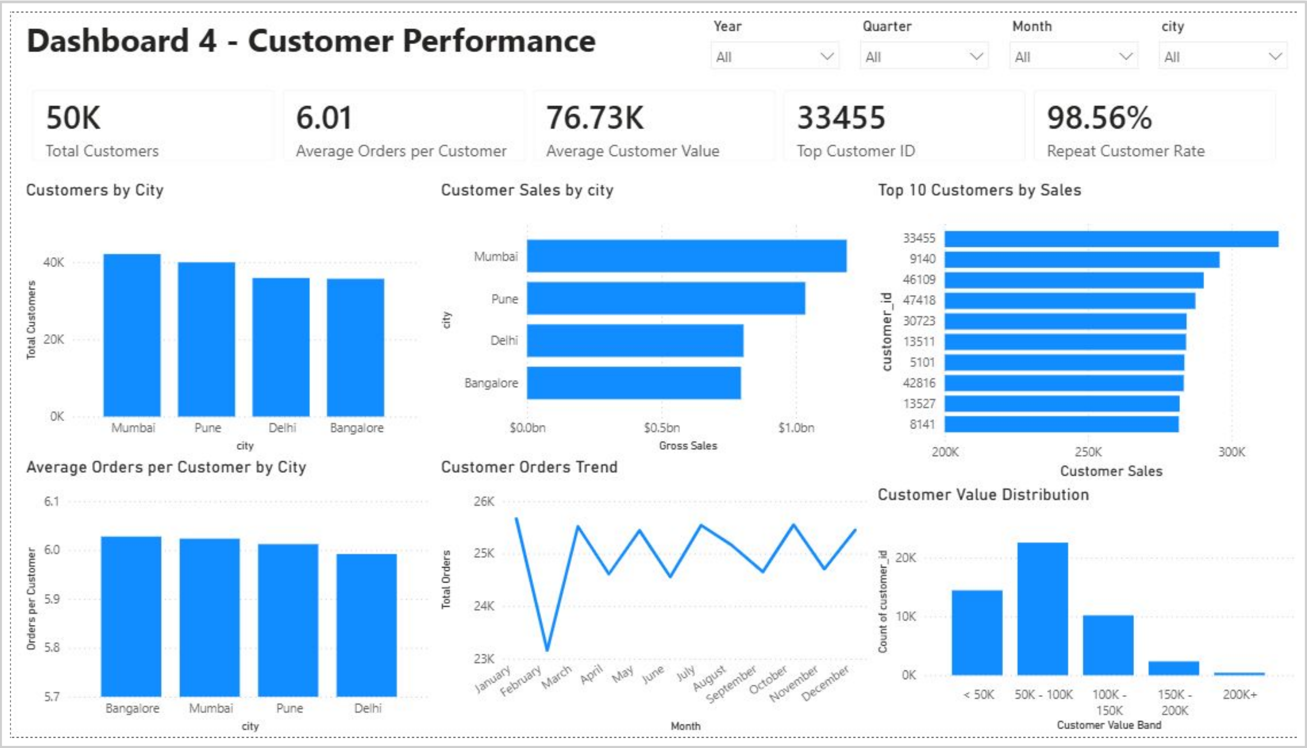
KEY FINDINGS

- Product 745 is the top product.
- Cat_5 is the top category.
- Approximately 10K distinct products are represented.
- Approximately 2M units are represented.
- Category AOV varies, creating cross-selling opportunities.

MANAGEMENT CONCLUSION

Product and category performance is relatively diversified, with Cat_5 leading and Product 745 as the top seller. Differences in category AOV indicate opportunities to optimize product mix, pricing and cross-selling.

Customer Performance



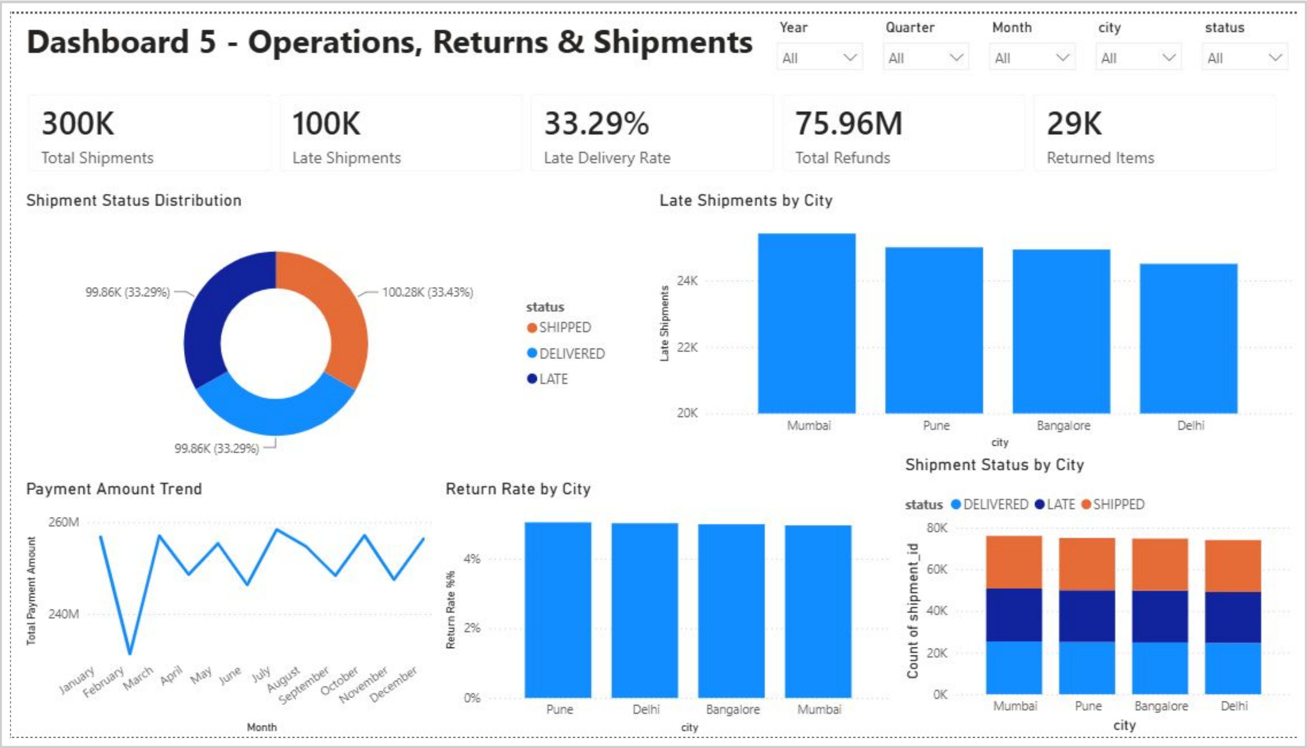
KEY FINDINGS

- Mumbai has the largest customer base and highest customer sales.
- Average orders per customer are around 6.
- Repeat Customer Rate is 98.56%.
- Largest customer-value segment is 50K–100K.
- High-value customers are a smaller segment.

MANAGEMENT CONCLUSION

Mumbai leads in customer base and sales value, with an exceptionally strong 98.56% repeat customer rate. The largest growth lever is increasing the value of existing customers rather than pure acquisition.

Operations, Returns & Shipments



KEY FINDINGS

- 99,855 of 300,000 shipments are late — a 33.29% late-delivery rate.
- Mumbai has the highest absolute late-shipment count.
- Pune has the highest return rate at 5.04%.
- Refunds total ₹75.96M.
- 29,251 order items were returned.

MANAGEMENT CONCLUSION

Operations analysis reveals a significant delivery challenge at a 33.29% late-delivery rate. Mumbai records the highest absolute late shipments while Pune has the highest return rate. Delivery reliability and return reduction are key operational priorities.

VALIDATION

Consolidated Validated KPI Register

All major Power BI KPIs and analytical outputs were reconciled against SQL before finalization.

Dashboard	Metric	Value
D1 Executive	Gross Sales	~₹3.83B
D1 Executive	YoY Growth	29.77%
D2 Store	Top Market	Mumbai (~31%)
D3 Product	Top Product / Category	745 / Cat_5
D4 Customer	Repeat Rate	98.56%
D4 Customer	Avg Customer Value	₹76.73K
D5 Operations	Late Delivery Rate	33.29%
D5 Operations	Refunds	₹75.96M

ACTION PLAN

Business Recommendations

1 Improve Delivery Reliability
High

Reduce the 33.29% late-delivery rate via monthly targets and bottleneck analysis.

2 Increase Customer Value
High

Leverage 98.56% repeat rate with loyalty, cross-sell and upsell programs.

3 Reduce Avoidable Returns
Medium

Investigate high-return products, categories and cities to cut refund cost.

4 Strengthen Market Performance
Medium

Sustain Mumbai's lead; replicate its strategies in Pune, Delhi, Bangalore.

5 Optimize Product Portfolio
Medium

Identify high/low performers among ~10K products for SKU rationalization.

6 Leverage Q4 Performance
Low

Apply Q4 demand patterns to inventory & promotions in weaker quarters.

ANALYTICAL INTEGRITY

Data Limitations & Caveats

SOURCE-DATA LIMITATIONS

- No explicit product cost/COGS — profit cannot be derived.
- Signup-date inconsistency affects tenure/cohort analysis.
- No return-date field — only order-month association.
- Multiple return records can reference one order item.
- No payment method/status fields available.
- Shipment status limited to SHIPPED, DELIVERED, LATE.

PREFERRED TERMINOLOGY

Use	Avoid
Gross sales / revenue	Profit / margin
Promotion-associated sales	Sales caused by promotion
Distinct returned order items	Return records = unique items
Payment difference	Payment error
Higher return rate among late orders	Late shipment caused the return

Thank You

Multi-Branch Retail Performance & Operations Intelligence

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